

Use of the PV-RT Power System Cyberattack Benchmark Platform

This benchmark platform was developed to compare the robustness performance of different controllers under cyberattack conditions in a PV-integrated two-area load frequency control (LFC) system in a fair and reproducible manner. In this platform, each controller is tested under the same load variation and cyberattack scenarios. Therefore, the obtained results can be directly compared based on controller performance.

The main scenario file used in the benchmark platform is `CyberAttack_Benchmark_PV_RT_Power_System.xlsx`. In this file, each row represents a normal operating or cyberattack scenario to be tested. The ID column represents the scenario number. The SLP2 column represents the load change in the second area. The FDI1 and FDI2 columns represent the false data injection (FDI) attack coefficients for the first and second areas, respectively. The DoS1 and DoS2 columns represent the delay values applied in the denial-of-service (DoS) attack for the first and second areas, respectively. By using this scenario file, all controllers are simulated under the same SLP2, FDI1, FDI2, DoS1, and DoS2 values.

The main purpose of this structure is to allow researchers to test their own controllers under the same benchmark scenarios. To use the benchmark platform, the researcher only needs to replace the existing controller block in the `PV_RT_PI_FDI_DoS_2024a.slx` Simulink file with their own controller structure. Then, after defining the required controller parameters in the MATLAB workspace or in a separate parameter file, the `CyberAttack_Performance` script is executed. As a result, the ITAE-based performance results of the tested controller are obtained for all benchmark scenarios.

CyberAttack_Performance: Generation of Scenario-Based ITAE Results

The `CyberAttack_Performance` script runs all scenarios defined in the `CyberAttack_Benchmark_PV_RT_Power_System.xlsx` file sequentially. For each scenario, the corresponding SLP2, FDI1, FDI2, DoS1, and DoS2 values are assigned to the MATLAB workspace, and the PV-RT LFC cyberattack benchmark Simulink model is executed.

As a result of each simulation, the ITAE performance index is obtained. Since ITAE is a cumulative performance signal accumulated over time, the ITAE value reached at the end of the simulation is used for scenario-level performance evaluation. In the code, this value is obtained as `max(ITAE)`. Because the ITAE signal increases cumulatively, this value practically represents the total time-weighted error effect at the end of the simulation. A lower ITAE value indicates that the controller suppresses the frequency deviations and tie-line power deviation more quickly and more effectively.

After running the CyberAttack_Performance script, the ITAE values obtained for all scenarios are saved into an Excel file. For example, for the PI controller, the output file can be named as:

CyberAttack_ITAE_Results_PI.xlsx

Similarly, after replacing the controller block in the Simulink benchmark model with their own controller structure, researchers can run the CyberAttack_Performance script to generate the ITAE result file for their own controller.

CyberAttack_Performance_Results_Compare: Comparison of Controller Performances

The CyberAttack_Performance_Results_Compare script reads the ITAE result files obtained by the CyberAttack_Performance script and performs a comparative performance analysis between different controllers. In the provided example structure, the results of the proposed FOPPI controller can be compared with the results of the classical PPI or PI controller.

In this comparison, the file CyberAttack_ITAE_Results_FOPPI.xlsx represents the reference benchmark results of the Fractional-Order Predictive PI (FOPPI) controller proposed in the study entitled “Cyberattack-Resilient Fractional-Order PPI Control for Frequency Regulation in PV-Integrated Power Systems.” Researchers may use this file as a reference result and compare the performance of their own controllers with the FOPPI controller under the same cyberattack scenarios.

After generating the result file for their own controller using the CyberAttack_Performance script, researchers can define this file in the CyberAttack_Performance_Results_Compare script. In this way, the tested controller and the FOPPI controller can be directly compared under the same benchmark scenarios consisting of identical SLP2, FDI1, FDI2, DoS1, and DoS2 values. Performing the comparison using the same scenarios makes the evaluation fair and reproducible.

The CyberAttack_Performance_Results_Compare script calculates basic statistical metrics for the compared controllers. These metrics include the mean, standard deviation, median, minimum, and maximum ITAE values. In addition, histogram plots are generated to analyze the distribution of ITAE values, and a boxplot is created to visually compare the overall performance difference between the controllers.

General Usage Procedure

- 1) Open the PV_RT_PI_FDI_DoS_2024a.slx Simulink file.
- 2) Replace the existing PI controller block in the model with the controller structure developed by the researcher.

- 3) Define the required controller parameters in the MATLAB script or in a separate parameter file.
- 4) Run the CyberAttack_Performance script to generate ITAE results for all scenarios in the CyberAttack_Benchmark_PV_RT_Power_System.xlsx file.
- 5) Save the obtained results into an appropriate Excel file.
- 6) Define the generated result file in the CyberAttack_Performance_Results_Compare script.
- 7) Compare the performance of the tested controller with the proposed FOPPI controller or other reference controllers.

With this structure, different controllers can be tested under identical load variation and cyberattack scenarios. Therefore, the robustness performance of newly proposed or existing controllers under FDI and DoS attacks in PV-integrated two-area LFC systems can be evaluated in a fair and reproducible manner by using the FOPPI controller results as a reference benchmark.